LEARNING PORTFOLIO

NORDWIND SOLUTIONS

NAME OF GROUP MEMBERS:

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1. IMPACT BUSINESS MODEL CANVAS

Problem Statement

Buildings in Lappeenranta need local renewable energy; solar has seasonal limits and urban wind needs site-specific assessment

Mission Statement

Provide building-integrated wind systems with site analysis, installation, monitoring, and maintenance.

Key Partners

- Structural engineers
- Electrical installers
- Installation contractors
- Monitoring technology providers
- Housing companies and pilot customers
- University advisors
- Municipal/public property partners
- Funders and financiers

Key Activities

- Site analysis
- Design
- Sales
- Installation
- Monitoring
- Maintenance.

Key Resources

Team expertise, turbine concept, partner network, monitoring tools, pilot access.

Value Proposition

- Local renewable electricity generation
- Reduced grid dependence
- Site-specific building solutions
- Technically credible deployment
- Monitoring and maintenance support
- Visible sustainability action
- Complementary solution to solar

Stakeholder Relationships

- Consultative
- Transparent
- long-term service based.

Channels

- Direct outreach
- Meetings
- Networks
- Pilot project discussions
- Website
- Brochures and marketing materials
- Referrals and partnerships

Stakeholder Segments

Primary customers

- Housing companies
- Student housing providers
- Assisted-living organizations
- Public buildings
- University facilities
- Large residential property owners

Beneficiaries

- Building owners
- Facility managers
- Residents and building users
- Local community

Cost Structure

Hardware, installation, monitoring, maintenance, marketing, admin, technical support.

Revenue Streams

System sales, installation, maintenance, monitoring, consultation, upgrades.

Intended Impact

More local renewable energy, lower grid dependence, lower emissions, stronger urban renewable adoption.

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2. INTERVIEWS FOR BUSINESS IDEA VALIDATION

- **Interview 1**

- **Interviewee:** Ville Varma
  Occupation: Property Manager
  LOAS (Student Housing Foundation for the Region of Lappeenranta)
  Date: 17.03.2026
  Method / place: Email

LUT Group Confidential - Other information (3Y)

Topics discussed:

- Building energy needs in student housing
- Interest in local renewable-electricity generation
- Concerns related to rooftop or facade wind-turbine systems
- Importance of maintenance, monitoring, and technical reliability
- Factors influencing adoption decisions

Key Findings:

- Cost is an important factor in any decision related to adopting a new building energy solution.
- Maintenance must be simple and practical, because complex upkeep would reduce the attractiveness of the solution.
- Proof of performance is necessary before adoption, meaning that the customer would need credible evidence that the system works effectively in real building conditions.

- The interviewee saw potential in the business idea and expressed interest in the service.
- They are willing to pay if cost, maintenance simplicity, and proof of performance are satisfactory.

- **Interview 2**

- **Interviewee:** Petri Kuukka
  Occupation: Systems specialist
  Aspakoti housing
  Date: 19.03.2026
  Method / place: Email

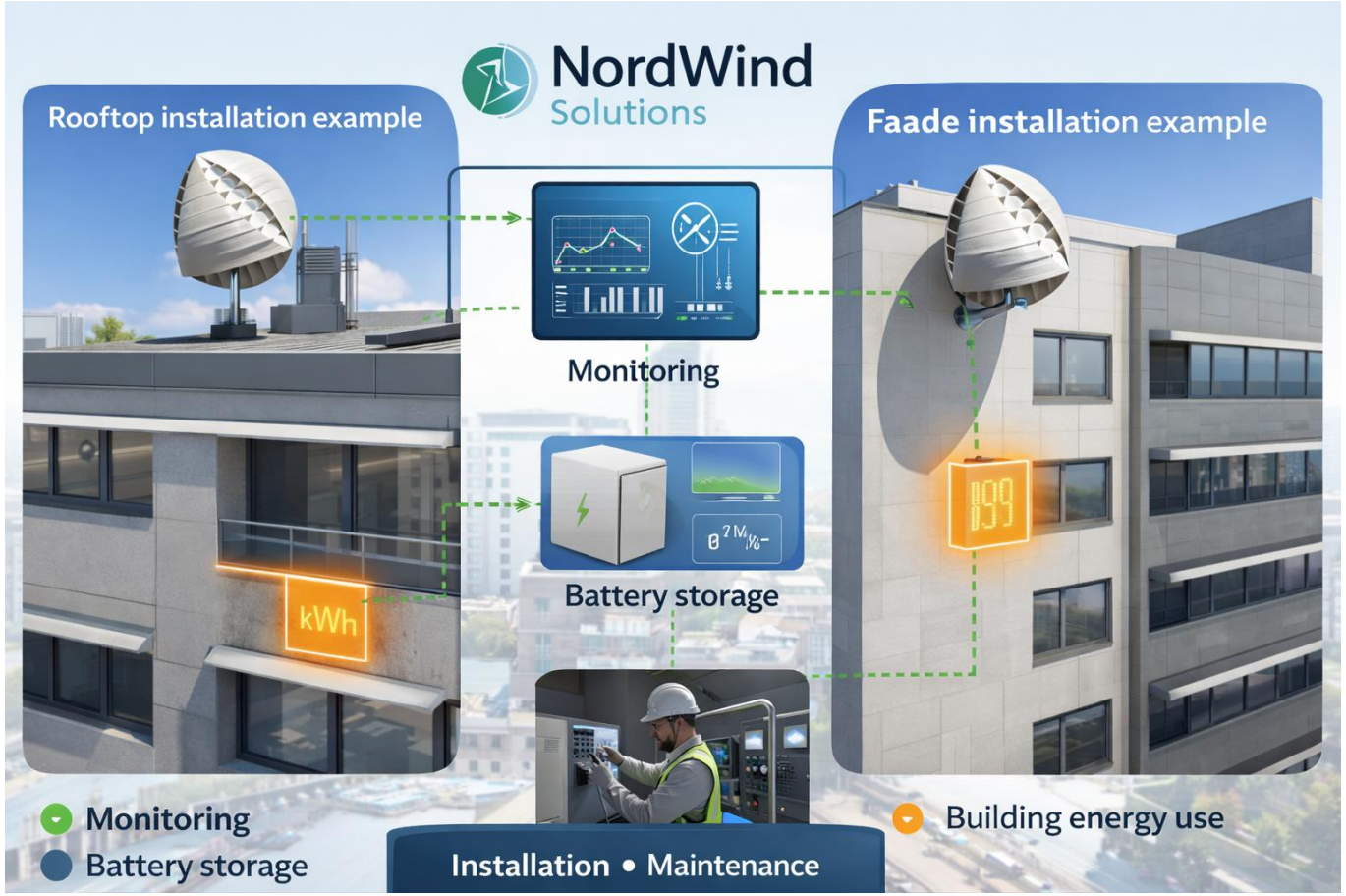
Topics discussed:

- Suitability of urban wind solutions in Finland
- Role of local renewable-electricity generation in buildings
- Importance of technical feasibility and performance validation
- Integration of wind systems with building energy use
- Long-term sustainability relevance of the concept

Key Findings:

- Technical feasibility and building-specific assessment are important for this type of solution.
- Performance validation is necessary before wider adoption.
- The concept is more credible when presented as part of a broader building energy strategy rather than as a stand-alone claim.

•The interviewee indicated that this type of product/service is relevant and needed in suitable building contexts.
•The interviewee showed willingness to pay for a credible, validated solution with clear technical evidence and practical value.

3. DIGITAL PROTOTYPE

Digital prototype of the NordWind Solutions building-integrated omnidirectional wind turbine system is made with help of AI.